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CONTRIBUTORS

Ellen Mantus, Rapporteur; Committee to Identify Innovative Technologies to Advance Pharmaceutical Manufacturing; Board on Chemical Sciences and Technology; Division on Earth and Life Studies; National Academies of Sciences, Engineering, and Medicine

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Proceedings of a Workshop

IN BRIEF

MAY 2020

Innovations in Pharmaceutical Manufacturing

Proceedings of a Workshop—in Brief

Global pandemics and the increasing severity and frequency of natural disasters have highlighted the vulnerabilities of drug supply chains and have underscored the need to modernize pharmaceutical manufacturing. The workshop Innovations in Pharmaceutical Manufacturing held in Washington, DC, on February 27–28, 2020, provided a venue for discussing potential technologies that are on the horizon in the next 5–10 years in the pharmaceutical industry. It was hosted by the National Academies of Sciences, Engineering, and Medicine Committee to Identify Innovative Technologies to Advance Pharmaceutical Manufacturing and served as its first information-gathering activity for this committee.¹ This Proceedings of a Workshop—in Brief summarizes the presentations and discussions that took place during the workshop. The workshop videos and presentations are available online.²

REFLECTIONS FROM THE CENTER FOR DRUG EVALUATION AND RESEARCH

Janet Woodcock, director of the US Food and Drug Administration (FDA) Center for Drug Evaluation and Research (CDER), opened the workshop by providing an agency perspective. Almost 2 decades ago, CDER launched an initiative called Pharmaceutical Quality for the 21st Century with the goal of achieving an agile, flexible pharmaceutical manufacturing sector that reliably produces high-quality drugs without the need for extensive regulatory oversight. She noted that advanced manufacturing has moved from the laboratory feasibility stage to commercial applications, that innovative technologies are in the pipeline, and that proposals are being submitted to the CDER Emerging Technologies Team from all types of companies throughout the pharmaceutical sector. CDER has also approved several continuous manufacturing applications. Woodcock stated that the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use is developing standards and attempting to incorporate new manufacturing techniques into the existing regulatory structures. Because pharmaceutical manufacturing is a global activity, she emphasized the need for all regulatory agencies to coordinate and cooperate and hoped that the industry could eventually move to a single global dossier, that is, a single regulatory evaluation.

Some progress has been made, but Woodcock stated that much more is needed, especially to achieve an agile, flexible manufacturing system. The coronavirus pandemic and recent natural disasters have exposed the fragility of the current drug supply chain. Lack of redundancy in manufacturing sites, the absence of surge capacity, and the complexity of supply chains create serious vulnerabilities in our drug supply, she said. Miniaturized, modular, and continuous production capacity now being developed might be able to address those deficiencies, but it will take years to commercialize it widely. She said that the emergence of novel, patient-focused, or individualized therapies also is putting pressure on the system to develop fit-for-purpose manufacturing and control strategies. As an aside, she noted that the pharmaceutical industry needs to adopt sophisticated control strategies similar to those of other industries. Such approaches enable real-time monitoring of product quality and might result in more efficient regulatory oversight. She emphasized the need for a new, highly trained, multidisciplinary workforce that is expert in the innovative technologies that are being developed for the pharmaceutical industry.

¹ For information on the committee's membership and task, see <https://www.nationalacademies.org/our-work/identifying-innovative-technologies-to-advance-pharmaceutical-manufacturing>.

² See <https://www.nationalacademies.org/event/02-27-2020/identifying-innovative-technologies-to-advance-pharmaceutical-manufacturing-workshop-1-and-meeting-2>.

Woodcock stated that the agency needs to prepare scientifically and technically for the innovative technologies that are being developed. She emphasized the need for advances in regulatory policy given that the agency is unsure how some of the innovative ideas will fit into the regulatory framework. Industry's reluctance to embrace new technologies, she said, is probably related to expected regulatory obstacles with FDA and other regulators, and promotion of broad adoption of advanced manufacturing will likely require incentives. To assist the agency, Woodcock stated, the committee has been asked to produce a consensus study report that will identify emerging technologies—such as product technologies, manufacturing processes, control and testing strategies, and platform technologies—that potentially could advance pharmaceutical quality and modernize manufacturing of products regulated by CDER. The committee was also asked to provide insights on technical and regulatory barriers to innovations and to provide recommendations for overcoming the barriers in its report. Woodcock closed by noting that the committee's input will be crucial in protecting the nation's drug supply.

DRUG PRODUCT MANUFACTURING

Daniel Blackwood, a research fellow in the Drug Product Design Group of Pharmaceutical Sciences—Small Molecule at Pfizer, began the session on drug product manufacturing by describing activities focused on continuous manufacturing in his company. He noted that interest in continuous manufacturing arose several decades ago as the industry began to prepare for patent expiry of its blockbuster, high-volume medicines. The focus on personalized medicine over the last decade has also underscored the need to innovate.

Blackwood stated that continuous manufacturing initiates a cascade of transformational advances in technology. It allows process intensification, which enables miniaturization of systems that have small footprints and reduced energy consumption. Miniaturization makes modularity and ultimately portability possible. Blackwood stated that focusing on portable, continuous, miniature, and modular technology will allow Pfizer to transform how it develops, manufactures, and distributes its drug products. Such technology might make it possible for pharmaceutical companies to share space and possibly some operations if precompetitive agreements are in place.

Blackwood next discussed Pfizer's experience with continuous direct compression in the manufacture of several solid oral drug products. Taking into consideration process-feed concerns, sensor limitations, content uniformity concerns, and challenges related to blend properties, its engineers selected two candidates with which to investigate the new approach. Using key capital investments, Pfizer was able to convert powders to film-coated tablets in minutes and accelerate approval or registration of some products. Blackwood noted that Pfizer has continued to investigate applications of the process to other products. He added that Pfizer also has made incremental innovations in existing technologies, for example, by incorporating sensors to monitor mixing processes and enabling integrated control with real-time decision-making. An incremental innovation envisioned is to reduce or simplify the elements in continuous feeding and mixing operations so that cleaning processes can be accelerated. To innovate further, Blackwood concluded, Pfizer is investing heavily in computational process modeling, digital design, transformation of big data into insight, and multiparticulate dosage forms, which provide flexibility in bringing medicines to diverse patient populations.

Govind Rao, a professor of chemical and biochemical engineering at the University of Maryland, Baltimore County, continued the discussion of drug product manufacturing by describing a portable, agile system for producing FDA-approved biologics in less than 24 hours. His focus is on reducing health care costs; current projections are not sustainable, and one approach to changing that is to decrease the cost of pharmaceuticals. The problem, however, is that it takes many years for an invention to become an innovation, and he hopes that the time can be reduced.

Rao stated that his system was created as a result of the vision of the Defense Advanced Research Projects Agency to develop compact, robust, automated systems for manufacturing biologics at the point of care within a few hours. His system uses a cell-free approach to produce proteins (Adiga et al. 2018). The advantage of his system is that it uses a lyophilized formulation that can be activated by adding buffer and DNA. RNA is then synthesized and translated into protein. The technology is consistent and can produce product rapidly, for example, in 2 hours. The software used is not complex; Rao noted that it does not take sophisticated software, for example, to read a chromatograph or to monitor output from various sensors. He has demonstrated the system's potential by synthesizing purified envelope protein from the Zika virus in less than 24 hours and by producing the monoclonal-antibody drug Humira (adalimumab). As a first step in obtaining FDA approval, Rao and his team conducted a demonstration project with Neupogen (filgrastim), a well-understood drug that already had FDA approval, and showed that their system was safe and efficacious (Adiga et al. 2020).

Rao mentioned a project in collaboration with GE Healthcare that has led to the development of new remote bioreactor sensors that can be used to monitor analytes, oxygen, and carbon dioxide in small-scale systems in real time. In response to an audience question, Rao noted that it is hoped that the sensors will provide better monitoring and better process information and that they are being evaluated on different scales and in various reactor types. The exciting aspect, Rao concluded, is the application of the sensors in the medical field, for example, as transdermal sensors to monitor premature babies, patients in clinical trials, or simply people who take drugs to treat various conditions.

Akhilesh Bhambhani, head of the New Technologies–Vaccine Drug Product Development Group at Merck, described two advances in drug product production at his company. He stated that the problem concerning innovation is not the lack of ideas but the difficulties encountered in execution. He noted that there are two types of innovation: incremental, which tends to be short-term improvements in an existing market, and disruptive, which tends to be long-term fundamental changes or breakthroughs in new markets. He added that the drivers for innovation depend on whether it is in a high-income or low-income market but that improving human health worldwide will require an integrated drug product strategy and that the ultimate goal should be to improve the affordability and accessibility of products.

One advance in drug product production has been the development of lyosphere technology in which a dried drug product—a vaccine or biologic—is produced as a consistent bead, Bhambhani said. The approach is advantageous in that it reduces the space needed for storage, allows various products to be combined easily, enables titration of doses, and allows flexible packaging to be used. The product can be improved by using high-disaccharide formulations; for example, using these formulations can greatly improve product stability. He added that Merck has also developed lyospheres for targeted oral delivery by adding a coating that resists disintegration until it reaches the target site.

Another advance that Bhambhani described was the development of microwave vacuum drying that achieves dehydration at lower temperatures. He noted that drying is faster than lyophilization because heat transfer occurs by radiation rather than conduction. He listed several advantages: (1) faster drying technology enables semicontinuous manufacturing, (2) the technology is compatible with multiple delivery devices, (3) one can achieve enhanced thermostability by using high-disaccharide formulations, and (4) the technology has a smaller footprint and lower operating costs than current lyophilization processes. Bhambhani emphasized the reduction in drying time with this technology—typically from days to hours. He said that his company wants to create flexible manufacturing with new technology that has the ability to produce small product batches. To achieve that goal, Merck will rely on portable manufacturing units, robotics to improve compliance, data analytics and information technology integration, and continuous manufacturing.

Richard Korsmeyer, president of Korsmeyer Consulting and former executive director of Advanced External Projects in Pharmaceutical Sciences at Pfizer, discussed innovations in dosage forms. He noted that much enabling technology exists but that the key is to identify applications that justify sophisticated approaches. Over the years, people have wondered whether it is possible to develop a “magic bullet,” but Korsmeyer countered that the question might be whether there is a “magic target” inasmuch as many of the cell-surface proteins that serve as targets are present in almost all cells. Some, however, have posited that nanoparticles could serve as magic bullets.

Intense interest in using nanoparticles has arisen because they are small, are versatile, and can be designed to be multifunctional, Korsmeyer said. Several nanoparticles have been approved, and several more are in clinical trials. “Old” drugs have also been reformulated with nanotechnology to improve safety, although none has been shown to be more effective than the original product. Major challenges in using nanoparticles are associated with their complexity—there are multiple engineered components, multiple disciplines are needed for their design and development, their manufacture will probably be expensive, and their performance will depend on “hitting the target” precisely. Korsmeyer noted, however, that better understanding of cellular mechanisms and barriers should enable more deliberate design of functionalized nanoparticles, and there is an opportunity to engineer small-scale processes for early study that can be scaled up, for example, by using microfluidic approaches. Furthermore, imaging technologies and tools can help to determine whether nanoparticles are going where intended. The Nanotechnology Characterization Laboratory of the National Cancer Institute also can serve as a resource in the development of new nanotherapeutics.

In closing, Korsmeyer emphasized that nanoparticles are materials whose surfaces affect behavior and performance and that their characterization has implications for design and manufacture. He urged industry and regulatory authorities to be receptive to alternative manufacturing paradigms that could enable production of these challenging products.

David Lechuga-Ballesteros, a research fellow at AstraZeneca, discussed innovative formulations for biologics. He noted some of the challenges in delivery of biologics: many require refrigeration, they typically have poor physical stability and poor oral bioavailability, their portability often requires a temperature-controlled supply chain from production to delivery, and most are administered parenterally. However, drug delivery via microparticles has shown great promise for addressing many of those challenges.

Lechuga-Ballesteros stated that one way to create microparticles is by using spray drying and noted that low-density, spray-dried microparticles are ideal for pulmonary delivery. Inhaled antibiotics in powder form have been shown to be more efficacious and safer than intravenous delivery and can be delivered in lower doses and faster than nebulized-solution inhalation (Geller et al. 2011). Porous microparticles can also be designed to be effective carriers of crystalline active pharmaceutical ingredients (APIs), and cosuspensions have been shown to be more robust, consistent, and reliable than crystal-only suspensions (Doty et al. 2018). He emphasized that spray-dried microparticles have characteristics that make them ideal for pulmonary delivery and that proteins in the form of dry amorphous powders are stable at room temperature for more than 2 years. Another advantage is that the process is scalable, that is, from bench to commercial production.

Lechuga-Ballesteros described several other techniques for producing microparticles, including a microfabrication (“molding”) technique (Garcia et al. 2012) and a crystallization–freeze-drying process. He closed by stating that many innovative products for delivery of proteins and peptides are in the pipeline and briefly described pulmonary delivery for topical applications (Bodier-Montagutelli et al. 2018), nasal delivery of various products (Tiozzo Fasiolo et al. 2018), and oral delivery of proteins and peptides (Brown et al. 2020), which has posed a substantial challenge because of the harsh environment of the gastrointestinal tract.

Discussion

To close the drug product session, Matthew DeLisa, William L. Lewis Professor in the Robert Frederick Smith School of Chemical and Biomolecular Engineering at Cornell University, moderated a panel discussion with the speakers and the workshop audience. The speakers were asked to predict what FDA might see in the next 5–10 years and what technologies might be further in the future. Lechuga-Ballesteros proposed that oral delivery of peptides and proteins might be more common in the next few years and added that gene delivery through solid-lipid nanoparticles is probably years from commercialization. Korsmeyer stated that nanoparticle formulations that improve safety and tolerability of products will probably be common in the near future but that complex-nanoparticle formulations are not likely in the near term. He added that antibody-targeted particles and microfluidic processing might also be on the horizon. Bhambhani noted that a paradigm shift is needed to change how products are manufactured and delivered, and he emphasized the need for a manufacturing system that is agile and adaptable and that increases affordability and accessibility. He predicted new technologies to enable reduction in cost of goods. Rao predicted the entry of Amazon into the distributed manufacturing arena and added that patient privacy laws need to change.

Next, several speakers were asked to discuss barriers in advancing novel platforms or new technology. Bhambhani said that the mindset of “if it’s not broken, don’t fix it” is problematic. Another factor is that the market is constantly evolving, and this complicates assessment of the business case for change. A workshop participant commented that another barrier is competition with Amazon and Google for data scientists. Another participant asked Blackwood whether Pfizer has considered licensing the technology or the quality systems associated with its modular approach to ease barriers to other companies in adopting the approach. Blackwood noted that the technology is freely available to others and that Pfizer has been open regarding integrated control systems through publications and conference presentations. Korsmeyer noted that a cultural factor in the pharmaceutical industry affects its ability to innovate; the industry should consider what intellectual property it needs to own and control to run its business and be profitable. The drug, not the manufacturing process, should be protected or owned. Rao agreed and pointed to the computer industry as an example of standardization of components.

DeLisa closed the session by asking the speakers about lessons that they have learned regarding regulatory approval of innovative technologies. Lechuga-Ballesteros stated that there is more leeway today in changing a process if a product and its quality attributes are not changing. He emphasized that companies tend to be more conservative than FDA, which is willing to listen and make a decision on the basis of data. He added that large companies are becoming so process-oriented that it is difficult for new ideas to flourish. Blackwood stated that formation of the Emerging Technology Program in FDA has been extremely helpful in bringing new technologies to life and noted that in-depth, face-to-face discussions during preapproval inspection have been highly effective. Bhambhani highlighted the need to minimize the number of variables that change in a process to ease regulatory approval and the need to start discussions early, not only with regulatory groups but with manufacturing colleagues, to judge feasibility. Rao concluded that this is an industry in which no one wants to be the first to adopt a new technology; everyone wants to be second.

CONTROL AND ANALYTICS

Richard Braatz, Edwin R. Gilliland Professor of Chemical Engineering at the Massachusetts Institute of Technology, began the session on control and analytics by reviewing best practices in using models, data analytics, and machine learning in process development. He first listed several strategies to advance process development: (1) increase understanding and optimization of unit operations to enable process intensification, (2) automate high-throughput technology to accelerate research and development, (3) develop plug-and-play modules that have integrated control and monitoring to facilitate deployment, (4) develop dynamic models for unit operations that enable automated plantwide simulation and control design, and (5) exploit data analytics and machine learning. He noted that there are many applications of the strategies to pharmaceutical manufacturing and described one that uses an automated molecular synthesizer to produce, purify, and characterize a product by using flowsheet models, process intensification, optimized plug-and-play fluidic modules, and feedback control (Coley et al. 2019). He emphasized that all the strategies depend heavily on models.

Braatz stated that model development is not a linear process of making assumptions on the basis of process knowledge and then building a model by using process data. Rather, it is an iterative process in which models are refined until a

satisfactory measure of success is achieved. He emphasized that selecting the best data-analytics tool is difficult and requires substantial expertise. There are many tools, and users typically apply the tools that they know, and this can have suboptimal results. A systematic approach to tool selection, however, allows a user to focus on objectives rather than on methods. Braatz offered two approaches to managing data and tool selection. One is to run a variety of models and then select the best one by minimizing observed cross-validation error. He noted that evaluating too many models can be problematic and that the key is to select a few that are known to be suitable for the data and the type of application. The second approach is simply to select the best model on the basis of the problem type and data characteristics, that is, assess the nonlinearity, multicollinearity, and dynamics of the data and then select the model that is best on the basis of the characteristics. He provided several examples of models that could be used given specific data characteristics.

In closing, Braatz predicted that there will be advances in methods that combine data analytics and machine learning with first-principles models, increased use of tensorial data streams, and a broadening of the scope of data analytics and machine learning in pharmaceutical applications. He imagined that the industry will automatically archive data for process modeling and analysis and improving monitoring and control, use plantwide data to optimize operational models, use all available data to ensure that product quality specifications are met, and ultimately use ecosystem data to connect customer needs to manufacturing data.

John Schiel, a research chemist in the Biomolecular Measurement Division of the National Institute of Standards and Technology (NIST), described analytical technologies for characterizing and monitoring therapeutic proteins, primarily monoclonal antibodies (mAbs). He noted the evolution of technologies from low to high resolution and said that technologies today can provide foundational knowledge on critical quality attributes (CQAs) that can be monitored to ensure safety and efficacy of biopharmaceuticals. He added that many technologies are available and that the goal is to replace some of the older technologies with newer ones to increase product knowledge, decrease cost, and increase speed of drug development. He cautioned, however, that one needs to evaluate a method's suitability for a given purpose and stated that NIST provides reference materials or standards to evaluate innovative technologies and ensure that they are being applied appropriately.

Schiel stated that the NISTmAb is a highly stable biopharmaceutical-grade mAb that is used to develop innovative technology and evaluate new analytical methods. He continued that its physicochemical and biophysical attributes have been exhaustively characterized and highlighted global interlaboratory comparisons and various technologies that have been evaluated with the NIST mAb. First, he described the multiattribute-method consortium in which mass spectrometry was used to evaluate industrywide new peak (peptide) detection performance metrics. The study showed that the approach provides a sensitive impurity test and that false positives and false negatives can be mitigated with proper controls. Second, he discussed an interlaboratory comparison project in which protein structural dynamics were measured by hydrogen–deuterium exchange mass spectrometry (HDX-MS). That study was the first determination of HDX-MS reproducibility and indicated potential areas for improvement. Third, he described an interlaboratory study whose purpose was to establish metrics for a two-dimensional nuclear magnetic resonance spectroscopy (2-D NMR) method. The study showed that 2D-NMR spectral fingerprinting is repeatable and reproducible and that peak position is a robust measurement. He noted that this method is a powerful analytical tool in that the dataset contains information about each amino acid and the conformational location of each amino acid in the protein. Lastly, Schiel noted various technologies that can be used to evaluate conformational ensembles (Castellanos et al. 2018). He stated that protein therapeutics do not have a static structure and that understanding their dynamic structure might shed light on how it influences, for example, stability, interactions, or dangerous immune responses. In closing, Schiel noted that the NIST interlaboratory studies and analytical research and development have targeted emerging technologies that are expected to have broader use in the next 5–10 years and help to optimize the process for manufacturing protein therapeutics.

Karen Balss, a scientific fellow in Advanced Technology and Technical Operations at Janssen Pharmaceuticals, continued the discussion of analytics by describing the promise of sensors in pharmaceutical manufacturing. She first reflected on the issuance of FDA guidance on process analytical technology³ in 2004 and noted that much progress has been made in this field. Balss stated that sensors are key components of process analytical technology and that the industry needs to move from univariate sensors to more complex ones that can provide information on chemical, biological, and physical attributes. She continued that advanced process control could someday combine advanced sensors with machine-learning methods and empirical data to predict yields, demonstrate quality control, and ultimately enhance productivity.

Balss discussed several examples of the use of sensors and modeling to improve the manufacturing process. She emphasized the need to have sensors in each unit operation in the process and the value of in-line sensors that can provide

³ FDA defines process analytical technology as “a system for designing, analyzing, and controlling manufacturing through timely measurement ... of critical quality and performance attributes of raw and in-process materials and processes, with the goal of ensuring final product quality” (FDA 2004, p. 4).

real-time data to models so that needed corrections can be made quickly. Many vendors sell sensors that can provide direct, chemical-specific, quantitative measurement of product attributes. She said, however, that the technology is not as mature in the biologics space and discussed how valuable advanced sensors could be in monitoring a bioreactor. The typical properties monitored in a bioreactor are temperature, pH, carbon dioxide, and dissolved oxygen, she said; however, being able to monitor the feed or the product would be extremely valuable, and sensors to do so are available. She noted mass spectrometry as an option but favored other spectroscopic techniques, such as Raman, given their lower cost and their ability to provide a unique fingerprint with no sample pretreatment. In closing, she emphasized the need to make the business case for investing in advanced process control and hoped that enhancing models with advanced sensor data to optimize feed control and ultimately product quality would be more common in the next 5–10 years.

Jack Prior, head of Manufacturing Science, Global Manufacturing Science and Technology at Sanofi, provided his view of the current state of process data analytics. He stated that collecting, managing, and analyzing data are becoming more challenging as the industry moves from simply describing what is happening to predicting and controlling what is going to happen. He continued that there are barriers to leveraging process data to improve operations—physical barriers that involve capabilities to measure, access, and organize data and organizational barriers that involve questions of trust and the desire to analyze and act on the data. He noted that trust means not only accepting that data are accurate but proving that they are robust enough to use for a particular application. He then described three realities of data analytics for biological processes. First, manufacturing data are not the same as data from a designed experiment, and one has to be careful not to equate correlation with causality. Furthermore, the sources of variation are often not contained in the data. Second, biological processes are nonlinear and time-variant, and the industry needs to move away from multivariate analyses. Third, although the industry typically does not generate “big data,” integration and analysis of industry data require investment.

Prior presented 24 technologies in the manufacturing innovation lifecycle and noted that where a technology falls in the innovation lifecycle for a specific company depends on which technologies the company needs to enable successful production of its products. He highlighted three technologies on the horizon: digital twins that encapsulate process knowledge in a real-time “twin” that monitors and predicts process behavior, mixing validation that uses computational fluid dynamics to examine the entire vessel and surpasses conventional mixing assessment, and data-lake systems that store all the raw data that can then be used for various applications. He continued by noting several barriers in various phases that are inhibiting innovation. In the exploration phase, one needs data engineers, modelers, and domain experts who have access to large quantities of the right data that can be matched to the right problem. In the industrialization phase, there needs to be a critical-mass market for vendor commercialization and a digital infrastructure that allows integration and validation. For initial filings, one needs the right fit for a critical product or process and enough lead time to build the innovation into early process development; a problem raised earlier in the workshop is the fear of being first with the risk of approval delays. In the commercialization phase, barriers include substantive investment in legacy platforms and the need for global regulatory acceptance. To accelerate digital innovation, Prior concluded, the industry needs to invest in digital infrastructure and integration, there needs to be a joint effort to advance agile software validation, and FDA innovation-promotion programs and efforts in global regulatory harmonization need to continue.

Jun Huang, director of the Process Monitoring, Automation, and Control Group of Pfizer Global Technology & Engineering, presented an analytics and digital technology roadmap to advance biomanufacturing. He first envisioned a future state in which manufacturing plants have central data repositories that connect all plant operations and the supply chain and that increase visibility and predictive capabilities. He described the digital plant-maturity model proposed by the BioPhorum Operations Group and emphasized the role of analytics in ultimately creating the “adaptive” plant.⁴ In achieving the vision, Huang said that it is important to accomplish incremental wins that demonstrate the value of the technology—think big, start small, and scale fast. Specifically, one first needs to assess compelling business needs and problems and align the innovative technologies with corporate strategies, priorities, and fundamental value propositions. Second, one needs to identify case studies that can demonstrate the success and value of the technologies. After successful demonstrations, one can scale up and can implement and replicate across the network.

Huang next discussed how to connect the disparate systems to form a unified whole. He suggested that the industry and FDA consider the Industrial Internet of Things to enable connectivity, that is, to build contemporary data infrastructure to aggregate data in a central location and to use distributed analytics to enable automation and data-driven decision-making at all levels. Achievement of connectivity, he said, opens the doors to many opportunities or capabilities, such as predictive maintenance, integrated scheduling and control, and real-time monitoring, control, and optimization. He noted two examples in which Pfizer leveraged real-time connectivity and data. In one, Pfizer replaced end-product testing with a soft sensor to reduce drying-cycle time and increase productivity; in the other, Pfizer used advanced process control for fully automating and controlling pH in a production process and thus reducing product variabilities. In closing, Huang

⁴ See <https://www.biophorum.com/wp-content/uploads/2017/05/Digital-Plant-Maturity-Model-White-Paper.pdf>.

emphasized that accelerating process development will require aggregating data into a central repository that can be accessed by various users in development and manufacturing and require development of the analytics to leverage those data. Those efforts will improve end-to-end visibility, process robustness, and productivity.

Discussion

Seongkyu Yoon, a professor in the Francis College of Engineering at the University of Massachusetts Lowell, and Saly Romero-Torres, senior manager of Advanced Data Analytics at Biogen, moderated a discussion with the speakers and audience. Romero-Torres opened by asking the speakers to elaborate on the business process for successfully implementing innovative technologies. Huang said that Pfizer uses a stage-gate process in which a project is divided into stages; at each stage, a decision is made as to whether to proceed. He added that the first step—understanding the problems and matching business needs to the technologies—is of paramount importance. Balss stated that Janssen uses a similar approach but emphasized the importance of creating stages that can be realized quickly and demonstrate a return on investment. It is important to remember that one will need to convince one's own company of the need to innovate, she said.

An audience member asked the speakers when they thought real-time release⁵ would become common. Prior stated that he hoped that a digital release process—one that eliminates the paperwork—would emerge soon. Huang stated that there were already examples of FDA approvals with real-time release but noted the complexity of that approach for the final product and suggested a modified version that focuses on particular CQAs in the process. Braatz agreed that moving step by step and focusing on specific CQAs was the most sensible course.

Regarding emerging innovations, another audience member asked the speakers to identify an innovation that FDA might see in the next 5–10 years. Prior suggested again the idea of digital release but noted the advance of soft sensors. Balss stated that FDA should see a targeted reduction in the amount of quality-control testing and possibly increased use of multiattribute mass spectrometry methods. As an aside, Prior noted that there are concerns regarding measurement of more process characteristics because it might require companies to meet more specifications in their processes and thus more opportunity for process failures. Schiel responded that one does not necessarily set specifications for every attribute when one uses a multiattribute method but rather monitors the critical attributes that are most relevant and uses detection of new peaks as a global process or product metric to ensure purity. He added that FDA will probably see various analytical techniques used in new ways in regulatory filings. Braatz noted serious problems with how some are using data analytics and predicted commercially available software in the next 5 years that incorporates more artificial intelligence in the data analytics so that users do not have to decide what method or statistic to use. Huang hoped that there would be continued innovation in the development of digital-twin technology. He noted that it is more than simply creating a virtual copy but rather having connectivity between the physical process system and the virtual copy so that the data can be used to make predictions and prescribe actions to optimize the process.

An audience member raised the idea of an industry consortium to share process data as a possibility for improving models or data analytics. Schiel noted that process data are intimately tied to product data and that the industry would therefore not be eager to share. He added that integrating data from 30 companies that use 30 analytical techniques—not all of them correct—would be extremely difficult without harmonization. Braatz noted that the discussion suggested that the data lake described earlier was not so mature in most pharmaceutical companies; Huang countered that individual companies are creating data lakes, that is, collecting data from multiple disparate sources, and progress depends on the company.

As a final question, an audience member asked the speakers what talents or skills their companies need. Huang said that his company is focused on recruiting data architects who can build data infrastructure or central repositories, data engineers who can transform or aggregate data into a suitable format, and data scientists who can build models and analyze data. He added that a business analyst who can understand business priorities and translate them into operations is important. Prior stated that the ideal is to find a data scientist who has specific domain knowledge; finding people who will be able to use the right tool for the right application is key. Balss emphasized the importance of taking advantage of internal talents and providing training opportunities for current staff rather than searching for talents outside. Regarding training, several speakers noted the need to develop new approaches, programs, or curricula to train scientists to meet the industry's future needs.

DRUG SUBSTANCE PRODUCTION

Salvatore Mascia, founder and chief executive officer of CONTINUUS Pharmaceuticals, began the session on drug substance production by describing the development of a compact factory that uses continuous manufacturing technology to produce drugs on demand. He noted that access to medications is affected by high costs and drug shortages. He attributed those

⁵ Real-time release is the use of process or other manufacturing data to ensure the quality of a product rather than, for example, testing the end product.

problems to batch production of pharmaceuticals, a long, fragmented, expensive process. His company has focused on creating an integrated continuous manufacturing solution that takes raw material, creates the desired API, purifies the API, and produces the final dosage form in a single system that can operate fully automatically 24 hours each day, 7 days per week. Part of the system is a solvent-recovery station that purifies, separates, and reuses solvents used in the production process in a closed-loop indoor system. He said that no intermediate or API is isolated from the integrated system, but various spectroscopic techniques are located at strategic points to monitor the process to ensure product quality. He noted that the system is a technology platform in which different modules or unit operations can be interchanged and combined; that is, the system is not designed to produce only one specific product but can manufacture various products.

Mascia stated that the company now has a fully operational pilot plant and has successfully demonstrated the system by producing a high-volume generic drug. In that case, it was able to meet all target production profiles and reduce costs by 30–50%, number of unit operations by 80–85%, solvent use by more than 60%, energy expenditures by 50–60%, facility footprint by about 90%, and lead time from months to less than 48 hours. He added that the technology will have a favorable effect throughout the supply chain, from development to manufacturing through sales and distribution and ultimately patient care. The business plan entails a three-pronged approach: (1) provide a specific technical or engineering solution using its technology to a pharmaceutical company, (2) provide contract development and manufacturing of specialty drugs, and (3) manufacture high-quality, low-cost generic drugs and distribute them directly to retail pharmacists and hospital pharmacies. The vision is to have a multisuite facility that has a dedicated single-product suite and other multiproduct flexible suites that have low and high capacity. Rather than building many multisuite facilities, the plan is to take the system to client-owned facilities. He said that the goal is to have a commercial product on the market in 3 years. He added that a strategic use of the multisuite facility would be to produce drugs critical to the United States. With an investment of \$90–130 million, he said, the company could have six strategically relevant drugs on the market in 3.5 years. In closing, Mascia emphasized that this technology could fundamentally change the drug distribution system to lower the cost of pharmaceuticals substantially.

Jean Tom, head of Chemical Process Development in Global Product Development and Supply at Bristol-Myers Squibb, provided an industry perspective on innovations in the manufacture of synthetic small-molecule drug substances. She noted that about 70 FDA approvals in 2019 involved “small” molecules and that many of the approvals involved complex molecules that are difficult to synthesize. She described the innovations in the context of the stages of process development—route invention, process invention, process characterization, and process validation—and noted challenges related to each. Regarding route invention, she said that the industry is investigating the use of photochemistry, electrochemistry, and biocatalysis. She noted that photochemistry and electrochemistry approaches use reaction and reactor-specific parameters that are new to regulatory authorities and entail complex descriptions. Biocatalysis offers the possibility of using multiple enzymes to synthesize an API from starting materials in a single vessel (Huffman et al. 2019), but this “cascade” biocatalysis entails several technical challenges, such as complexity of the reaction network, characterization of evolved enzymes, and process control, characterization, and robustness of the cascades. Tom noted that complex molecules, such as oligonucleotides and peptides, involve new regulatory challenges, including various issues associated with impurities and use of analytical methods different from those used to characterize traditional small molecules. As a final note on route invention, Tom mentioned co-processed APIs and the question, from a regulatory standpoint, of whether to treat them as drug substances or as drug-product intermediates.⁶

Regarding process invention, Tom acknowledged that innovations have been focused on continuous manufacturing and noted several challenges, including process-control complexities, lack of first-principle understanding, development of real-time feedback loops and predictive modeling, and uncertainties of regulatory requirements. Regarding process characterization, she said that easy access to computational power and open-source codes have enabled more advanced data analysis and modeling. She hoped that there would be consistent regulatory acceptance in data packages that contain more model-generated results as opposed to experimental results. As an example of data analytic advances, she highlighted Bayesian probabilistic modeling as a framework in which process data that are generated during the development stage can be used to predict risks and establish controls to meet quality and robustness expectations on product launch (Tabora et al. 2019). In the future, she hoped that automation and artificial intelligence could be used to streamline data packages for filing.

In closing, Tom provided a few thoughts on introducing new technology into manufacturing. She said that acceptance depends on the organization. Some are resistant to change because cost savings are seen as minimal, capacity is judged to be sufficient, and no incentives are in place to support innovation. She said that a substantive regulatory issue related to emerging technologies is that there is no unified regulatory authority whose expectations are clear and consistent. Although senior FDA leadership might support innovation, reviewers tend to be more conservative, and this means extra scrutiny and greater resistance to unfamiliar technologies. Those realities cause companies to be risk averse because they understand the

⁶ A co-processed API is a drug substance that contains the API and at least one noncovalently bonded nonactive component that changes the API’s physical properties to improve or enable drug product manufacturing.

challenges inherent in introducing innovative processes and do not want to compromise filing timelines. Tom concluded by saying that there needs to be a global approach to avoid being limited by the most conservative approach.

Günter Jagschies, principal consultant at Gemini BioProcessing and formerly with GE Healthcare Life Sciences, discussed manufacturing of biologics and innovations on the horizon. He began by listing several challenges for the biopharmaceutical industry and said that addressing them will require a focus on increasing facility output and process yield, creating a flexible system to produce a variety of drugs on various scales, simplifying operations, decreasing infrastructure cost, improving drug quality, and streamlining regulatory compliance. He provided an integrated bioprocessing overview and described changes that can be expected in 5–10 years at various stages. In upstream processing, the focus will be on improving productivity, for example, by using high-volume cell banks, high-density inoculations, and concentrated media and by reducing cell perfusion rates and impurities. In midstream processing, there will be a direct link between the reactor and the first downstream purification step; the centrifuge will become obsolete. In downstream processing, there will be improvements in product capture, for example, by using affinity membranes and better connections and simplifications in product purification so that intermediate storage capacity is not needed. As a result, the facility footprint will shrink. Jagschies also noted that on-demand buffer preparation will likely be integrated into the system; this will eliminate the need for buffer storage, again shrinking the facility footprint. In analytics and control, continuous operations will trigger the development of more CQA relevant in-process analytics, and real-time release will require better in-process analytics and models. Jagschies acknowledged that those advances raise the question of how to make the models and artificial intelligence transparent for regulatory review.

In closing, Jagschies offered some general forecasts related to the biopharmaceutical industry. He said that there will be more single-use technology with advanced control features, “plug-and-play” unit operations in modular facilities, benchtop-scale operations for many small-market therapeutics, and closed processing for hygienic operation on a small scale. He added that the need for storage to support operations will mostly disappear. Because facility scale and consequently cost are predicted to decrease, he said, regulators should be prepared for the entry of inexperienced manufacturers into the market. He added that regulators should be trying to understand the effects of high-productivity operations on CQAs given the certain trend toward continuous operations to increase productivity. As a final note, Jagschies emphasized the importance of controlling impurities produced by cells in culture to avoid having to remove them later in the process and the importance of matching the increased upstream productivity with improved downstream technology to purify the product.

Jon Coffman, director of Bioprocess Technology and Engineering at AstraZeneca, presented a framework for next-generation manufacturing in the biopharmaceutical industry and described what might be accomplished in the next 3–5 years. He said that the goal is to achieve integrated and continuous protein processing and hoped that an ecosystem would emerge in which companies could obtain off-the-shelf systems. Overall, the framework involves moving from stainless-steel bioreactors to single-use bioreactors, as Jagschies predicted, and enabling continuous harvest that results in a small downstream flow. Coffman noted that a benefit of the continuous process is that it results in healthier cells. He described various stages of the process and noted how several companies were implementing the next-generation manufacturing framework.

Coffman presented a process flow diagram that represented an industry “common denominator” for equipment and facility design and proposed a stepwise implementation strategy for companies that want to adopt continuous and integrated processing. The first step, he said, is to install an alternating tangential-flow filtration system that enables continuous harvest from the bioreactor. The second step is to install at least two columns for downstream processing. Coffman noted that two columns do not add much more complexity than one column but that three or more columns substantially increase cost and complexity. The columns can be run in parallel or in series; running in series changes the operation only slightly but might require a new virus-removal validation. He noted that a more complicated system was likely to be rare in the near future. The third step is to have sensors at the column outlets. Coffman also recommended sensors at the inlets to allow real-time modeling to provide assurance that the system is operating as intended. Mechanistic and machine-learning models will be needed because next-generation manufacturing will generate substantially more data than batch manufacturing, which cannot be analyzed by human operators, he said. A last step is to install in-line conditioning for buffer preparation; this must be implemented for operation on a commercial scale. As a final note, Coffman emphasized the importance of having a bioburden control strategy and highlighted a few technologies—downstream membrane absorbers and countercurrent and high-performance tangential-flow filtration—that could be seen possibly in 6 years.

Discussion

Timothy Charlebois, vice president of Technology and Innovation Strategy for BioTherapeutics Pharmaceutical Sciences at Pfizer, moderated a panel discussion with the session speakers; Gregg Nyberg, associate vice president of Biologics Process Research and Development at Merck Research Laboratories; Jorg Thommes, head of chemistry, manufacturing, and control at the Bill & Melinda Gates Medical Research Institute; and Andreas Bommarius, professor of chemical and biomolecular

engineering at the Georgia Institute of Technology. Charlebois first invited Nyberg, Thommes, and Bommarius to provide their reflections on the presentations and highlight opportunities for the industry.

Nyberg highlighted continuous manufacturing and noted that it is probably not going to decrease the cost of goods radically but will substantially change capital costs given the much lower costs of building a manufacturing facility. Other benefits of continuous manufacturing are that it allows companies to right-size manufacturing capacity and provides the ability to make a variety of molecules. He noted the importance of having plug-and-play capabilities with automated systems so that the process does not have to be redesigned for every new product. Given earlier statements about intellectual-property concerns, he commented that the manufacture of biologics differs from small-molecule manufacture in that the process is valued as intellectual property. He concluded by emphasizing the need to continue searching for alternative hosts or methods for producing biologics.

Thommes stated that his organization focuses on global health and thus large patient populations who live under resource-limited conditions. Given that perspective and especially the need to supply solutions at a cost that can be borne by low- and middle-income economies, he emphasized that both the large, fixed infrastructure and the modular, continuously operated manufacturing systems might yield potential solutions for addressing global-health needs for biologics. He said that high-dose applications for large patient populations are probably best served through large-scale manufacturing, and low-dose applications for smaller, more localized patient populations might be better addressed through modular approaches. He added that the productivity improvements that can be made with advanced process controls are essential to make possible a truly industrialized supply of pharmaceuticals. He agreed with Nyberg that the industry needs to continue pursuing alternative expression systems for making biologics and suggested that the industry might learn some valuable lessons from other industries, such as ones that produce industrial enzymes.

Bommarius stated that he typically teaches his students that quality and cost are the two most important factors in pharmaceutical manufacturing but that the workshop has demonstrated to him that the two most important factors are drug shortages and the complexity of the supply chain. In light of the current coronavirus pandemic, the question of production location is probably going to become much more important. He noted that there is often a disconnect between academic research and industry needs and emphasized that more sustained funding in fields that are relevant to industry might facilitate solutions to drug shortages. He agreed with Thommes that the pharmaceutical industry could learn from other industries and highlighted continuous manufacturing and process intensification that have been achieved in other industries as possibly valuable examples from which to learn.

On conclusion of the opening remarks, Charlebois prompted the audience for comments or questions. An audience member asked the panel how to implement major improvements so that drug prices could decrease to a level affordable by the global population. Thommes said that an investment should be made in commercial manufacturing in the early stages of a drug's development. Bommarius noted that there is little recognition for improving, for example, productivity of an antibody production process; therefore, fundamental changes are needed in the research landscape in the United States and elsewhere. Coffman added that one must identify the perceived need 80% of the time and then build the technology to address it; only 20% of the time can one try to build technology because it is the right thing to do. He added that a company's financial structure often motivates the funding for technology changes; saving money in one unit might allow spending in another. Tom reminded the audience that there is a fundamental difference between biologics and small molecules; even small changes in the cost of goods is important in small-molecule manufacture.

Next, an audience member probed the panel on advances in model development, implementation, and capabilities. Nyberg noted that requirements or levels of scrutiny will depend on how a model is being used. It is one thing to use a model to try to understand a process and another to use a model to make processing decisions. Although every model is different, he said, it would be helpful to have a standardized approach for establishing when a model is appropriate for a specific context. Romero-Torres commented that the challenge depends on what system is being modeled and its variance. Systems that have many interdependent variables and high variance could be complex to model because many variables need to be measured to make a good prediction or estimation. The problem is that in many cases the variables cannot all be measured, so a good and reliable estimation about, for example, a particular quality attribute is not possible. In the near future, she thought, predictions in small-molecule manufacturing would become reliable but predictions in large-molecule manufacturing would be more challenging given the many variables that will need to be measured. In response to a complaint that models just become more and more complicated, Bommarius countered that the key is to identify the critical process parameters; once they have been identified, a model can be simplified.

In closing, the panelists and audience discussed the need to create integrated, “end-to-end” processes. Bhambani commented that there is no gain if an API is continuously manufactured and is then stored to be batch processed later into the drug product. Mascia added that “end to end” should encompass the path from raw material to delivery to patients. Tom emphasized that there are challenges given the different regulatory requirements for various drug substances and products and mentioned again the issues that her company has faced with co-processed APIs. As a final note, Mascia raised the

prospects of release testing only for the final drug product in a continuous manufacturing process; he asked, If the API is not isolated, why should a company have to release it for testing?

WHAT'S MISSING

Gintaras Reklaitis, Gedge Distinguished Professor of Chemical Engineering and professor of industrial and physical pharmacy at Purdue University, closed the workshop by moderating a discussion to identify important topics or issues that had not yet been raised. Katherine Tyner, associate director for science in the Office of Pharmaceutical Quality at FDA, noted that there have been about 1,000 submissions to FDA for nanomaterial drug products whose designs range from simple to complex and that are in different stages of development. She provided two references (D'Mello et al. 2017; themed issue of *The AAPS Journal*⁷) that she hoped the committee would review. One audience member encouraged the committee to look for the disruptive technology inasmuch as many of the technologies that have been discussed over the last 2 days are focused on improving current processes. Another audience member suggested that the committee investigate advances in aseptic manufacturing given the importance of sterility assurance in parenteral drugs.

Several audience members raised issues associated with the current regulatory framework. One is how to view portable manufacturing—when you move a unit from one address to another, does it become a “new” facility in the new location? Today, facilities are regulated by street address, not “license plate.” Another issue is related to on-demand manufacturing that can produce unique dose combinations of various pharmaceuticals; regulations are not designed for variable-dose combinations. A third is related to hybrid products—ones that blur the lines between what has traditionally been considered a small molecule and a large molecule; regulatory compliance for these new types of products can be unclear.

Several audience members debated issues associated with data analytics and control. Braatz noted that digital twins constitute a technology that the committee will likely discuss in its report and encouraged the committee to define the term because there is much confusion about what it means. An audience member noted the resistance to using multiple sensors to collect more data and posed the question, What happens when two sensors that measure the same parameter disagree? It leads to confusion as to which to trust or how to manage conflicting data. Coffman agreed that it is difficult to introduce new controls into a system unless there is a recognized benefit in productivity or control of product quality. He focuses on model-based control systems that would accomplish what would be impossible for a human being to achieve today. Romero-Torres commented that the goal of collecting more information on a process is to develop a model that can achieve the desired process control.

An audience member questioned whether substantially increased productivity in the context of biologics production will come from implementing some control measure or from better understanding of cell physiology. Romero-Torres commented that control strategies are aimed not so much at *improving* productivity or quality substantially as at *maintaining* productivity and desired quality. An audience member noted that identifying inhibitory metabolites through transcriptomics and metabolomics and better sensing technologies might lead to increases in productivity, but large gains will likely result only from the use of alternative hosts. Nyberg noted that publicly funded investment is needed in this arena; everyone uses Chinese hamster ovary cells because an ecosystem or infrastructure has been built around them, and this makes successful development of alternative hosts difficult.

In closing, Reklaitis reminded the audience to participate in the workshop on technical and regulatory barriers scheduled for June 2–3, 2020, and noted that the committee will use the information from this workshop and the June workshop to write its report, which is due in early 2021.

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⁷ See https://link.springer.com/journal/12248/topicalCollection/AC_d31a4b0d2a9cdc71d616967defca50c9.

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